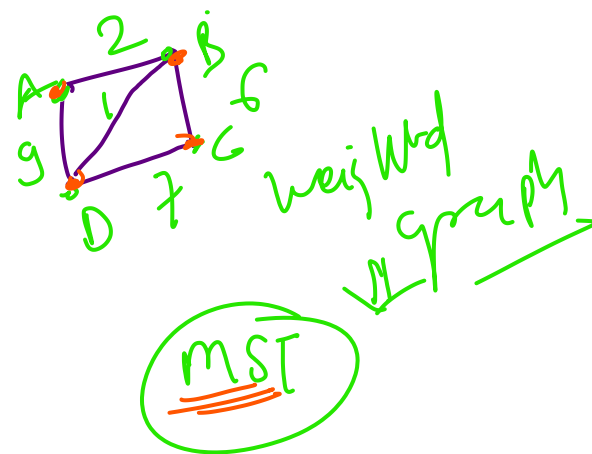
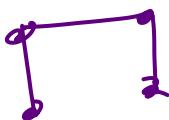
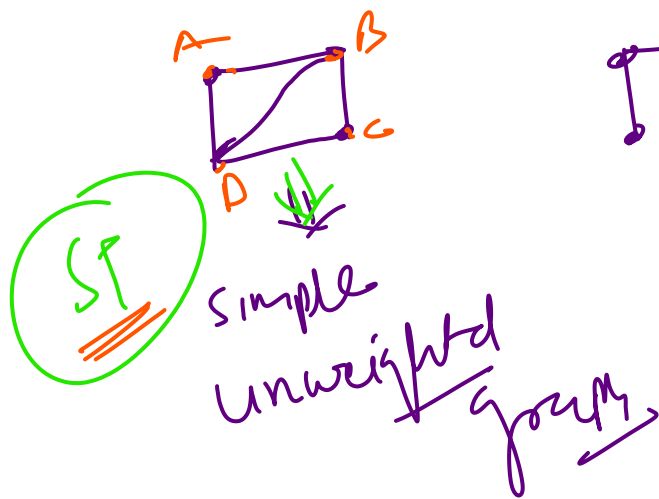


→ graph $\subseteq \mathbb{E}^n$
 $ST \subseteq \mathbb{E}^{n-1}$



Lec_65

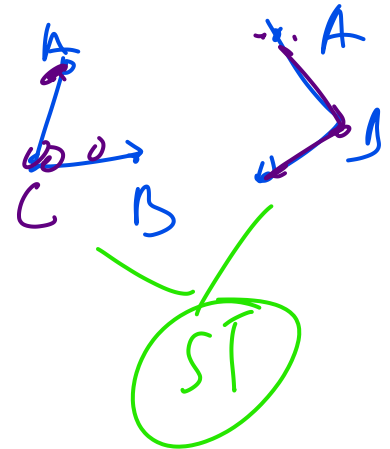
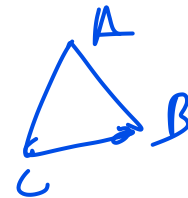
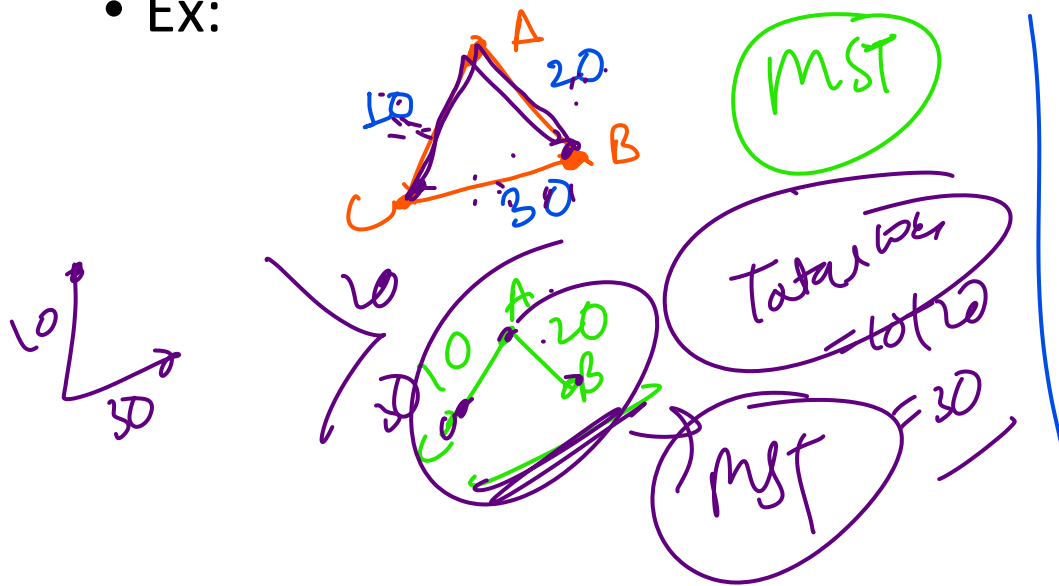
Minimum Spanning tree(MST)



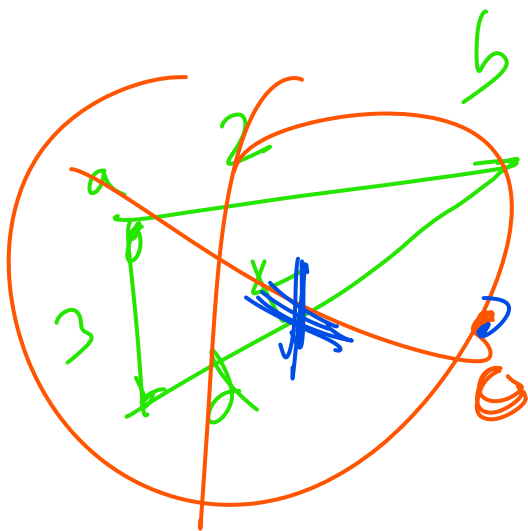
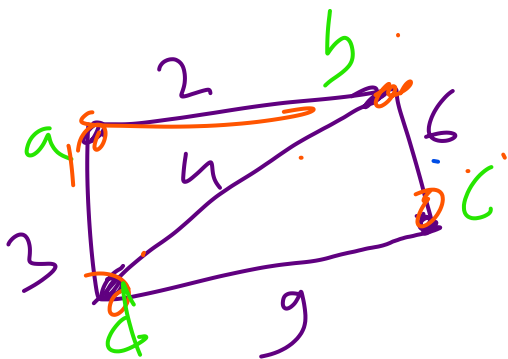
Minimum Spanning tree:

v vertex
 v_1 edge

- The minimum spanning tree has all the properties of a spanning tree with an added constraint of having the minimum possible weights among all possible spanning trees. Like a spanning tree
- Ex:



Ex:



graph

$$MST = 2$$

$$vertices = 4$$

$$edges = 5$$

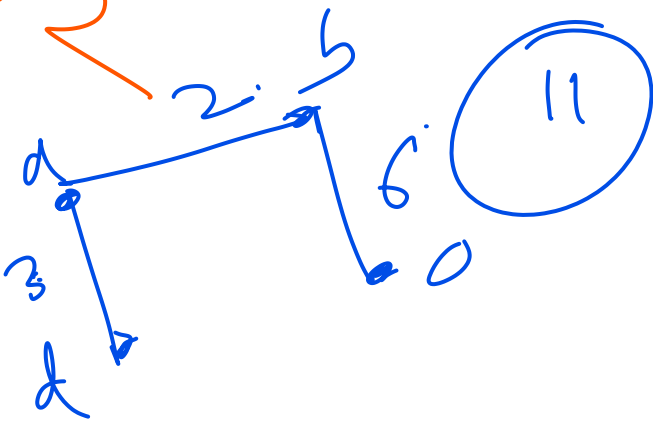
$$MST = 2$$

$$v = 4$$

$$e = 4 - 1$$

$$= 3$$

weight



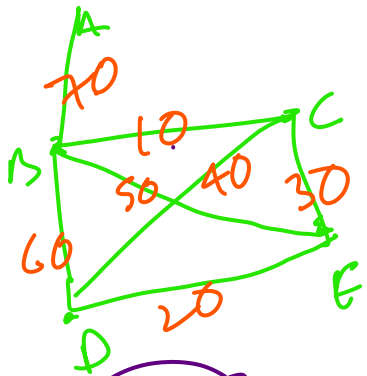
Algorithms for MST: to find mst we have 2 algo

- Kruskal's Algorithm
- Prims

In V vertex graph, $\text{vertices in } ST = V$
 $\text{edges " " } = V-1$

Kruskal's Algorithm: gives priority to edges

• Ex1:



$$\begin{aligned} V &= 5 \\ E &= 4 \end{aligned}$$

using Prim's algo find number of
 $mst = 7$.

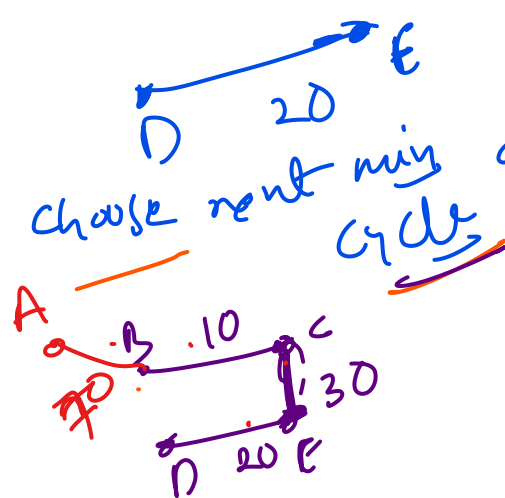
Step 1: - choose min weight edge

choose 2nd min weight edge

Step 2:

choose next min cycle

Step 3:

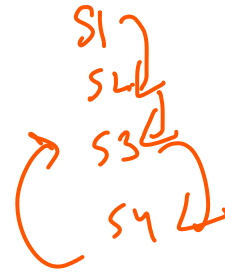
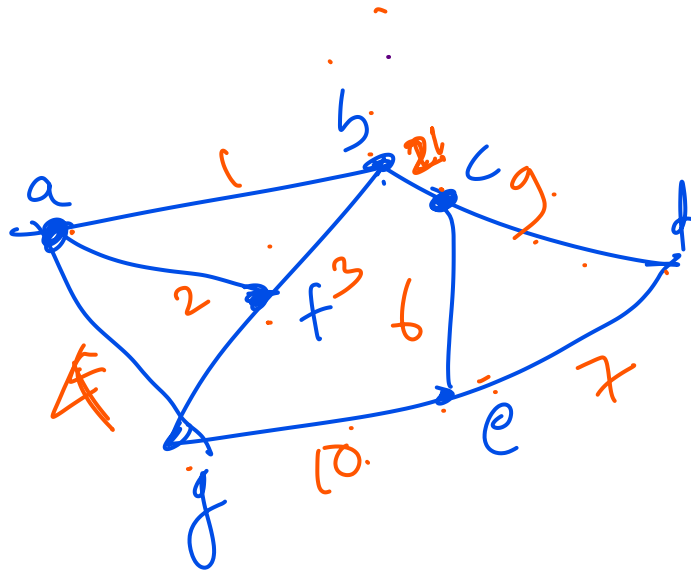


$$W = 70 + 10 + 20 + 30$$

and add if it is not forming
 $max = V, V-1$
 V vertex, $V-1$ edges

Step 4: repeat step 3 till $V-1$ edges

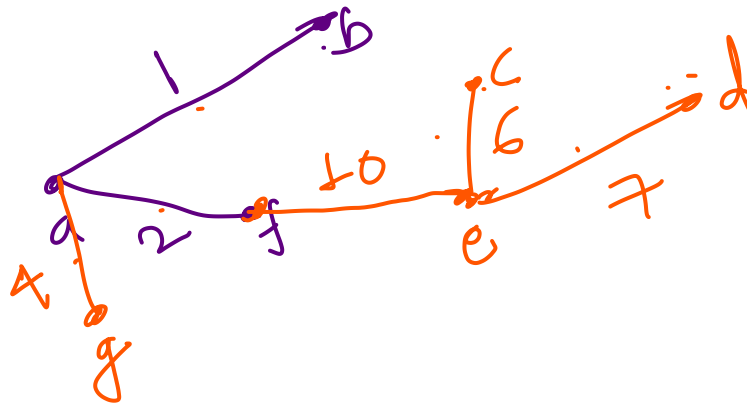
Ex2:



$$V = 7$$

$$E = 6$$

$$W = 2.$$

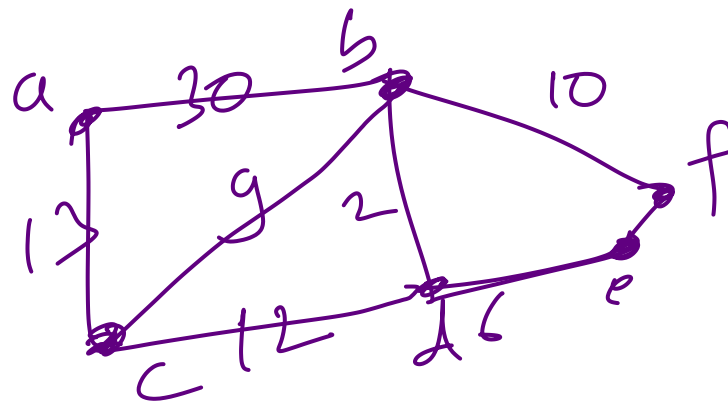


• min heap is used

Algorithm and TC:

∴ $S1 = \text{chosen}$
 $S2 = m$

min element in $O(1) \Rightarrow \text{minHeap}$



• made min heap of the edges (weight).

↳ ans  let E edges $\Rightarrow O(E)$ $\Rightarrow$ using build heap method

• delete min edge in each step $\Rightarrow O(\log E)$

• we will take $V-1$ edges

- 1 create min heap $O(E \log E) = O(E^2)$
- 2 delete $\downarrow$ min edges one by one = $O(\log E)$

#3 repeat till $V-1$ edges.

BC $\Rightarrow$ if in between no cycle
 then we need to delete $V-1$ edges.

WC $\Rightarrow$ if each time we are making cycle
 we have to take E edges.

$T_C = ?$

BC
 $O(E) + (V-1) \log E$

$\Rightarrow E + V \log E$

$E + V \log E$

WG
 $O(E) + E \log E$

$E + E \log E$

$E \log E$

~~W~~

V

